

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Project Setup	USN-1	As a developer, I can set up the Node.js + Express.js project with MVC folders and a MongoDB (Mongoose) connection.	3	High	Tharun Kumar S
Sprint-1		USN-2	As a developer, I can define the User schema with validation (name, unique email, hashed password, role).	2	High	Tharun Kumar S
Sprint-1	Registration	USN-3	As a user, I can register by entering my name, email and password.	3	High	Velan P
Sprint-1		USN-4	As a user, my password is encrypted with bcryptjs before it is stored.	2	High	Velan P
Sprint-1	Login	USN-5	As a user, I can log in with email and password and receive a JWT token.	3	High	Velan P
Sprint-1		USN-6	As a user, only authenticated requests can reach protected routes (JWT middleware).	2	High	Tharun Kumar S
Sprint-1	Study Material	USN-7	As a student, I can upload or enter study material (title, subject, content).	3	High	Varun Krishna S N
Sprint-1		USN-8	As a student, I can view and delete my saved study materials.	2	Medium	Varun Krishna S N
Sprint-2	AI Summary	USN-9	As a student, I can integrate the Google Gemini API through a dedicated AI service layer.	5	High	Vignesh B

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2		USN-10	As a student, I can generate a concise AI summary from my study material.	3	High	Varun Krishna S N
Sprint-2		USN-11	As a student, my generated summary is saved and linked to the material.	2	High	Vignesh B
Sprint-2	Flashcards	USN-12	As a student, I can generate flashcards (question and answer) from my material.	3	High	Varun Krishna S N
Sprint-2		USN-13	As a student, I can view my saved flashcards.	2	Medium	Vignesh B
Sprint-2	Quiz	USN-14	As a student, I can generate a multiple-choice quiz from my material.	3	High	Varun Krishna S N
Sprint-2		USN-15	As a student, my quiz is saved so I can retake it later.	2	Medium	Tharun Kumar S
Sprint-3	Study Plan	USN-16	As a student, I can enter my exam date and available time to get a personalised AI study plan.	5	High	Varun Krishna S N
Sprint-3		USN-17	As a student, I can view and save my study plans.	2	Medium	Tharun Kumar S
Sprint-3	Dashboard	USN-18	As a student, I can see all my previously generated resources in one place.	3	Medium	Vignesh B
Sprint-3		USN-19	As a student, I can use a React dashboard to upload material and access AI features.	5	Medium	Vignesh B
Sprint-3	Role-Based Access	USN-20	As an admin, my role gives me access to admin-only APIs that students cannot use (RBAC).	3	High	Velan P
Sprint-3		USN-21	As an admin, I can view and manage registered users.	2	Medium	Velan P
Sprint-4	Admin Monitoring	USN-22	As an admin, I can monitor AI usage and review system logs.	3	Medium	Velan P
Sprint-4	Testing	USN-23	As a tester, I can verify all APIs with Postman (positive and negative cases).	3	High	Vignesh B
Sprint-4		USN-24	As a team, we complete User Acceptance Testing and fix reported bugs.	5	High	Vignesh B
Sprint-4	Security & Validation	USN-25	As a developer, I add input validation, error handling and rate limiting on AI routes.	3	Medium	Tharun Kumar S

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	Deployment	USN-26	As a team, we deploy the backend and connect a hosted MongoDB database.	4	High	Velan P
Sprint-4	Documentation	USN-27	As a team, we finish the project documentation and demo.	2	Low	Vignesh B

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	4 Days	15 Sep 2026	18 Sep 2026	20	18 Sep 2026
Sprint-2	20	4 Days	19 Sep 2026	22 Sep 2026	20	22 Sep 2026
Sprint-3	20	4 Days	23 Sep 2026	26 Sep 2026	20	26 Sep 2026
Sprint-4	20	4 Days	27 Sep 2026	30 Sep 2026	20	30 Sep 2026

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

Project velocity: Our team (4 members) plans 20 story points per sprint and each sprint is 6 days long, so the team velocity is 20 story points per sprint.

Average velocity (AV) = story points per sprint ÷ sprint duration = $20 \div 6 \approx 3.33$ story points per day.

Total planned work = 27 user stories = 80 story points, delivered over 4 sprints (24 days, 5 Oct 2026 – 31 Oct 2026).

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

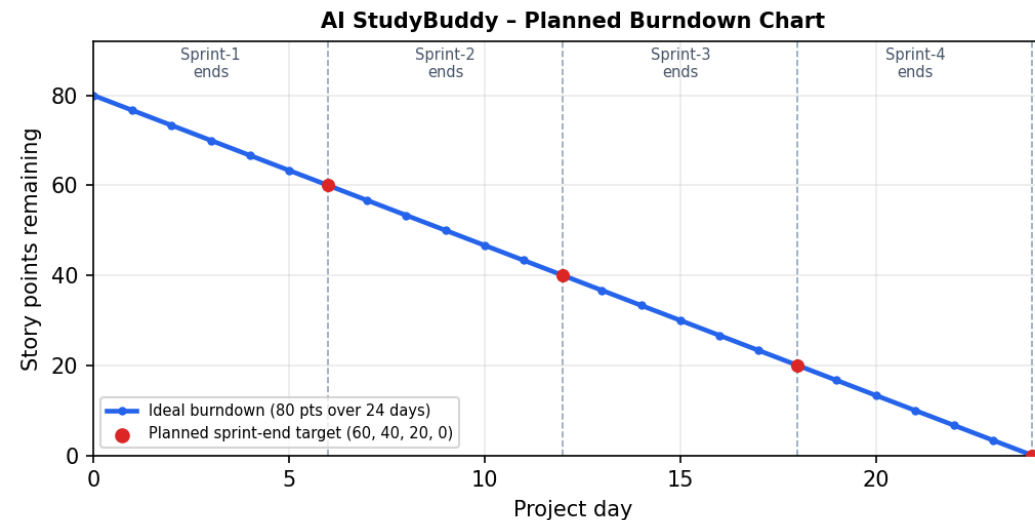


Figure 1: Planned (ideal) burndown for 80 story points across 4 sprints. Actual remaining points are added after each sprint from the Project Tracker.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>
<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>
<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>
<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Scrum Guide: <https://scrumguides.org/scrum-guide.html>